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Lab Database System

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Lab 04: Normalization

Scenario Hospital Patient Visit

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1. Introduction

This lab focuses on transforming an unnormalized hospital database into a well-structured relational schema using normalization techniques up to **Third Normal Form (3NF)**.

The given scenario represents hospital patient visits where all data (patient, doctor, department, diagnosis, and fee) is initially stored in a single table. This leads to redundancy and anomalies.

The objective is to:

- Identify **functional dependencies**
- Convert data into **1NF, 2NF, and 3NF**
- Eliminate anomalies
- Ensure **data integrity and consistency**

2. Problem Description

The hospital maintains records of:

- Patient details
- Doctor and department details
- Visit diagnosis and fee

Initially, all data is stored in one large table (**UNF - Unnormalized Form**), causing:

- Data redundancy
- Update issues
- Poor structure

3. Functional Dependencies

From the given dataset:

Primary Functional Dependencies

- VisitID → VisitDate, PatientID, DoctorID, Diagnosis, Fee
- PatientID → PatientName, PatientPhone
- DoctorID → DoctorName, Specialty

- Specialty → DeptName, DeptHead

Derived Dependencies

- VisitID → PatientName, PatientPhone
- VisitID → DoctorName, Specialty, DeptName, DeptHead

Candidate Key

- **VisitID** (uniquely identifies each record)

4. Unnormalized Form (UNF)

In UNF, all attributes are stored in a single table:

Problems:

- Redundant data (e.g., patient info repeated)
- Difficult updates
- Poor scalability

◆ UNF Table

In UNF, all information is stored in one large table without proper structure

VisitID	VisitDate	PatientID	PatientName	PatientPhone	DoctorID	DoctorName
V-9001	4/10/2026	P-201	Hassan	0300-1112233	D-30	Dr. Imran
V-9002	4/10/2026	P-202	Mehreen	0301-4445566	D-31	Dr. Asma
V-9003	4/11/2026	P-201	Hassan	0300-1112233	D-31	Dr. Asma
V-9004	4/12/2026	P-203	Junaid	0302-7778899	D-30	Dr. Imran

VisitID	Specialty	DeptName	DeptHead	Diagnosis	Fee (PKR)
V-9001	Cardiology	Heart Care	Dr. Tariq	Hypertension	2500
V-9002	Dermatology	Skin Clinic	Dr. Asma	Eczema	2000
V-9003	Dermatology	Skin Clinic	Dr. Asma	Allergy	2000
V-9004	Cardiology	Heart Care	Dr. Tariq	Arrhythmia	3000

5. First Normal Form (1NF)

A table is in 1NF if:

- All attributes are atomic
- No repeating groups exist

Implementation

The table visit_1nf ensures:

- Each field contains a single value
- Each row represents one visit

Observation

Although atomicity is achieved, redundancy still exists:

- Patient data repeats
- Doctor and department data repeats

The table already contains atomic values, so it satisfies 1NF after defining a primary key.

6. Second Normal Form (2NF)

A table is in 2NF if:

- It is in 1NF
- No partial dependency exists

Process

Since the primary key is **VisitID (single attribute)**, partial dependency is not an issue. However, logical separation is applied to remove redundancy:

Decomposition

- **Patient Table**

  PatientID → PatientName, PatientPhone

- **Doctor Table**

  DoctorID → DoctorName, Specialty

- **Visit Table**

  VisitID → VisitDate, PatientID, DoctorID, Diagnosis,

Result

- Reduced redundancy
- Improved data organization

◆ 2NF Tables

1. Patients Table:

PatientID (PK)	PatientName	PatientPhone
P-201	Hassan	0300-1112233
P-202	Mehreen	0301-4445566
P-203	Junaid	0302-7778899

2. Doctors Table

DoctorID (PK)	DoctorName	Specialty
D-30	Dr. Imran	Cardiology
D-31	Dr. Asma	Dermatology
P-203	Junaid	0302-7778899

3. Visit Table

VisitID (PK)	VisitDate	PatientID (FK)	DoctorID (FK)	Diagnosis	Fee
V-9001	4/10/2026	P-201	D-30	Hypertension	2500
V-9002	4/10/2026	P-202	D-31	Eczema	2000
V-9003	4/11/2026	P-201	D-31	Allergy	2000
V-9004	4/12/2026	P-203	D-30	Arrhythmia	3000

7. Third Normal Form (3NF)

A table is in 3NF if:

- It is in 2NF

- No transitive dependency exists

Transitive Dependency Identified

- DoctorID → Specialty
- Specialty → DeptName, DeptHead

◆ 3NF Tables

1. Patients Table

PatientID (PK)	PatientName	PatientPhone
P-201	Hassan	0300-1112233
P-202	Mehreen	0301-4445566
P-203	Junaid	0302-7778899

2. Departments Table

DeptName (PK)	DeptHead
Heart Care	Dr. Tariq
Skin Clinic	Dr. Asma

3. Doctors Table

DoctorID (PK)	DoctorName	Specialty	DeptName (FK)
D-30	Dr. Imran	Cardiology	Heart Care
D-31	Dr. Asma	Dermatology	Skin Clinic

4. Visit Table

VisitID (PK)	VisitDate	PatientID (FK)	DoctorID (FK)	Diagnosis	Fee
V-9001	4/10/2026	P-201	D-30	Hypertension	2500
V-9002	4/10/2026	P-202	D-31	Eczema	2000
V-9003	4/11/2026	P-201	D-31	Allergy	2000
V-9004	4/12/2026	P-203	D-30	Arrhythmia	3000

SOLUTION

Separate department data:

Final 3NF Tables

1. **Patient**
    1. PatientID (PK)
    2. PatientName
    3. PatientPhone
2. **Department**
    1. DeptID (PK)
    2. DeptName
    3. DeptHead
3. **Doctor (3NF)**
    1. DoctorID (PK)
    2. DoctorName
    3. Specialty
    4. DeptID (FK)
4. **Visit (3NF)**
    1. VisitID (PK)
    2. VisitDate
    3. PatientID (FK)
    4. DoctorID (FK)
    5. Diagnosis
    6. Fee

Outcome

- No redundancy
- No transitive dependency
- Strong referential integrity

8. Final Query (Reconstruction of Original Table)

The original dataset can be recreated using JOIN operations:

```
SELECT
    v.VisitID,
    v.VisitDate,
    p.PatientID,
    p.PatientName,
    p.PatientPhone,
    d.DoctorID,
    d.DoctorName,
    d.Specialty,
    dept.DeptName,
    dept.DeptHead,
    v.Diagnosis,
    v.Fee
FROM visit_3nf v
JOIN patient p ON v.PatientID = p.PatientID
JOIN doctor_3nf d ON v.DoctorID = d.DoctorID
JOIN department dept ON d.DeptID = dept.DeptID;
```

9. Anomalies and Their Elimination

Before Normalization

1. **Insertion Anomaly**
    - Cannot insert doctor/department without visit
2. **Update Anomaly**
    - Updating doctor or department requires multiple changes
3. **Deletion Anomaly**
    - Deleting a visit removes important data

After Normalization (3NF)

1) Insertion anomaly removed
2) Update anomaly removed
3) Deletion anomaly removed

10. Conclusion

This lab demonstrates how normalization improves database design by:

- Eliminating redundancy
- Ensuring consistency
- Improving maintainability

The final 3NF schema provides:

- Efficient data storage
- Strong relationships using foreign keys
- Scalability for real-world hospital systems